MILITARY SPECIFICATION

HIGH VOLTAGE POWER SUPPLY: 11739861

1. SCOPE

1.1 This specification covers one type of electronic assembly known as High Voltage Power Supply, 11739861 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control, Materiel: General Specification Governing the Manufacture And Inspection Of High Voltage Power Supply
MIL-P-14232	Parts, Equipment And Tools For Army Materiel, Packaging And Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance, And Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communications, And Radar For Aerospace, And Control Systems, Procedures 4
MIL-S-46844	Solder Bath Soldering Of Printed Wiring Assemblies, Automatic Machine Type
MIL-P-46843	Printed Circuit Assemblies, Design And Production of
MIL-P-55640	Printed Wiring Boards, Multi-Layered (Plated-Through Hole)

STANDARDS

Military

MIL-STD-105	Sampling Procedures And Tables For inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-810	Environmental Test Methods
MS25252	Lamp, Neon, Midget Inverted Flanged Base

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. Army, Frankford Arsenal

F11739861	High Voltage Power Supply: F11739861

PACKAGING DATA SHEET

11739861	High Voltage Power Supply

(copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - This module shall be manufactured in accordance with Drawing F11739861 and all drawings pertaining thereto.

3.2 General specification. - The power supply shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Electrical and electronic assemblies.
(e) Part identification of marking.
(f) Workmanship.

3.2.1 Sub-assemblies and components.- The high voltage power supply shall be comprised of the following components which should have been inspected and accepted in accordance with their respective drawings and applicable documents prior to incorporating into the printed circuit board:

(a) Gasket, shielding	A11737797
(b) Diode	A11739810
(c) Capacitor	A11739817
(d) Resistor	A11739818
(e) Voltage regulator	A11739820
(f) Inductor filter	A11739821
(g) Inductor filter	A11839822
(h) Transformer	A11S739823
(i) Transistor	A11739838
(j) Sub-assembly	E11739855

3.2.1.1 Printed circuit assemblies.- Pri nted circuit assemblies shall be in accordance with MIL-P-46843 and MIL-P-55640.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

3.2.1.2 Flow soldering. - Flow soldering shall be in accordance with MIL-S-46844 and MIL-S-45743.

3.3 Performance.

3.3.1 Interchangeability.- The high voltage power supply module shall meet all of the requirements of this specification and shall be interchangeable with all other power supply modules such that the replacement of any other module shall in no way cause incompatibility with the requirements of this specification.

3.3.2 Environmental.

3.3.2.1 Temperature storage.- Module to meet requirements of 3.3.3 subsequent to being exposed and thermally stabilized at temperatures of -650 and +155° Fahrenheit (F).

3.3.2.2 Operating temperature.- Module to meet requirements of 3.3.3 while being exposed to temperatures of O°F to 100°F.

3.3.2.3 Vibration. - Module to be mounted as used. Test per MIL-STD-810, Method 514.2, Procedure VIII, Curve V, Figure 514.2-6 resonance dwell not required. Module shall meet requirements of 3.3.3 subsequent to this test.

3.3.2.4 Shock. - Test per MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-1, Note 1. Module to meet requirement of 3.3.3 subsequent to this test.

3.3.2.5 Humidity. - Test per MIL-STD-810, Method 507.1, Procedure IV, Figure 507.1-3, chamber temperature not to exceed 155°F. Module to meet requirement of 3.3.3 after this test.

3.3.2.6 Fungus. - Test per MIL-STD-810, Method 508.1, Procedure I. Power supply modules to meet requirements of 3.3.3 subsequent to this test.

3.3.3 Operation.

3.3.3.1 No load voltage. - When 24 ± 6 VDC is applied to Pin “J” of Connector “Pi, and Pins “L” and “E” are grounded, the output voltage measured between Connectors XA3P1 and XA3P2 shall be 9.0 ± 1.0K VDC. (Reference Figure 1). Adjust Resistor R106 as required. The power supply module neon lamp shall be illuminated.

3.3.3.2 Full load current. - When 24 + 6 VDC is applied to Pin “J” of connector P1, Pins “L” and “E” are grounded and a total resistance of 195.7K * 2% OHMS is connected in series with a 1 - 10 milliampere meter between Connectors XA3P1 and XA3p2, the full load current as read on the meter shall be 6 ± 0.3 milliamperes. The voltage measured between Pin “M” and Pin “A” of Connector “J1” shall be 95 ± 20 VDC when a neon lamp (MS 25252-C7A) is connected in series with a 68K ± 10% resistor across Pins “M” and “A”. The voltage measured between Pins “M” and “E” shall be 230 + 10 VDC (Reference Figure 1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

3.3.3.3 Inhibit voltage test.- When 24 ± 6 VDC is applied to Pin "J" of connector P1, Pin "L" and Pin "E" of P1 are grounded and a total resistance of 195.7K ± 2% OHMS is connected across XA3P1 and XA3P2, the voltage measured between Pin "A" and Pin "E" of Connector "J1" shall be 135 ± 10 VDC,

3.3.3.4 Flash mode test. - When Pin "E" of Connector "JI" is connected to ground and 24 ± 6 VDC is applied to Pin "J" for a period of 0.5 second; minimum and Pin "L" is simultaneously grounded for a period of 100 milliseconds from the gated output of Timing Module 11737703. The voltage waveform appearing at Pin "L" shall be as shown in Figure 2 and the voltage waveform appearing at Pin "A" shall be as shown in Figure 3. A total resistance of 195.7K ± 2% OHMS shall be connected across XA3P1 and XA3P2 during this test.

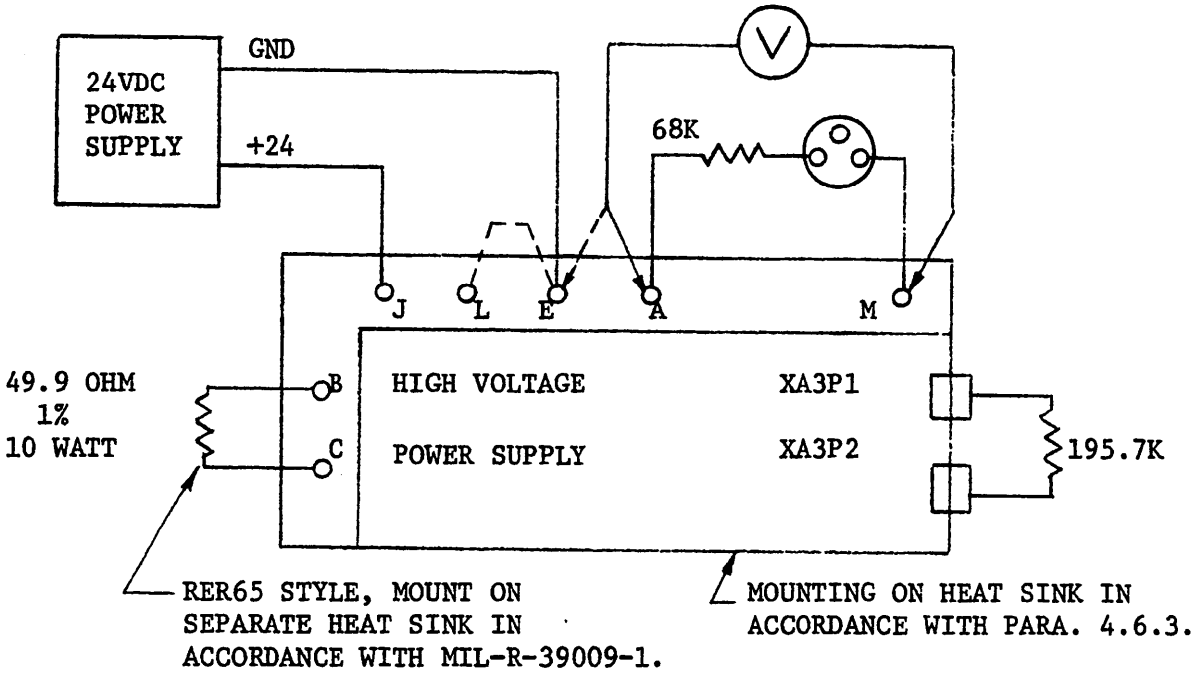


FIGURE I

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

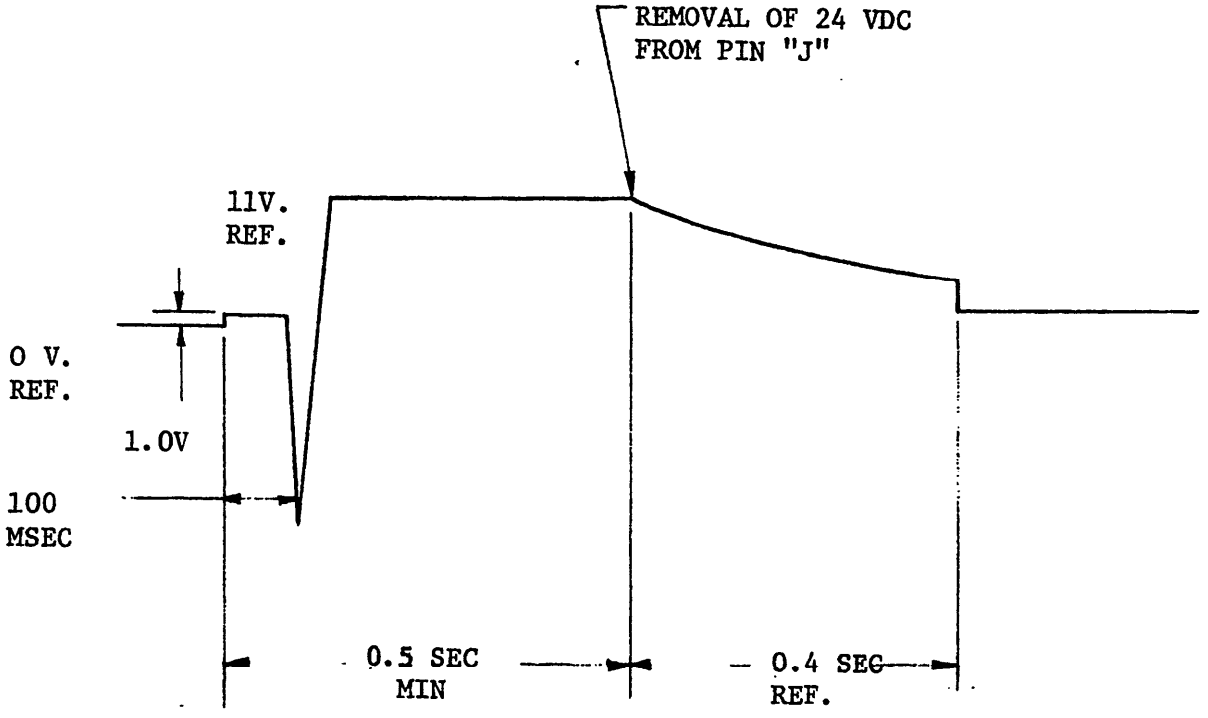


FIG. 2

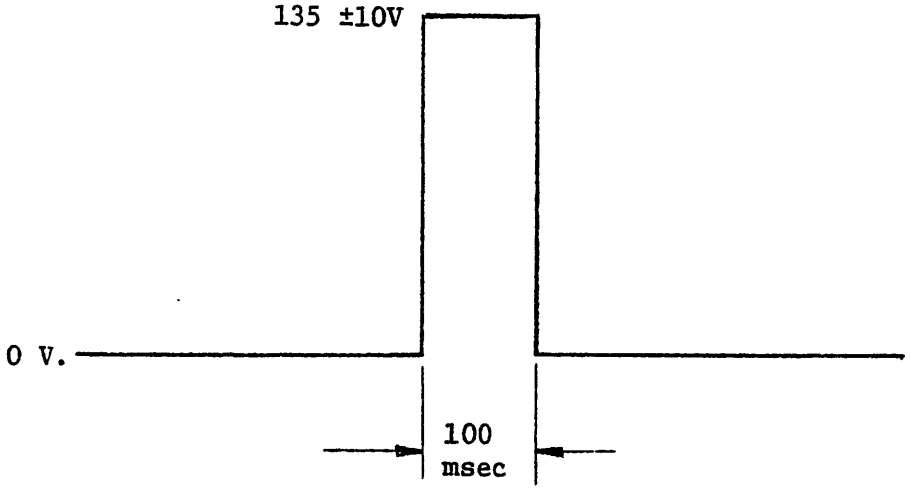


FIG. 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The three first article samples will be submitted to all the tests indicated in Table I and Table II. Of the test indicated in Table III and Table IV, one sample shall be subjected to the Fungus test (4.6.2.6) and the other samples shall be subjected to the other test specified.

4.2.1 Government test in.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contacting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-105.

4.3.2 Examination and tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.3.2.1 Components and subassemblies.- All compoents and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP's) listed in the Technical Data Package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and-tests-for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS
(100% INSPECTION)

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Interchangeability	3.3.1	4.6.1
102.	No load voltage	3.3.3.1	4.6.3.1
103.	Full load current	3.3.3.2	4.6.3.2
104.	Inhibit voltage test	3.3.3.3	4.6.3.3
105.	Flash mode test	3.3.3.4	4.6.3.4

NOTE: The tests in Table I shall be conducted at a temperature between +70°F and +80°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials manufacturing processes and techniques previously certified.

NO.	CHARACTERISTIC	TEST DATA TO COMPLY WITH
301.	Cleaning	MIL-P-46843
302.	Drying	"
303.	Coating	"

4.4 Special sampling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.4 Special sampling.

4.4.1 General.- Three assemblies shall be selected at random as a special sampling from each 50 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	PER DWG OR SPEC	TEST PROCEDURE
401.	Materials	3.1	Applicable dwgs & specs. - visual
402.	Fabrication	3.1	Applicable dwgs & specs. - visual
403.	General specification	3.2	MIL-F-13926 - visual

NOTE: The tests in Table II shall be conducted at a temperature between +70°F and +80°F.

4.4.2 Environmental. - Three assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	PER DWG OR SPEC	TEST PROCEDURE
404.	Storage temperature (-65°F to +155°F)	3.3.2.1	4.6.2.1
405.	Operating temperature (0°F to 100°F)	3.3.2.2	4.6.2.2
406.	Vibration	3.3.2.3	4.6.2.3
407.	Shock	3.3.2.4	4.6.2.4

4.4.2.1 Control tests.- Three assemblies shall be selected and meet the requirements and test of Table IV. These tests shall be conducted as follow-up tests to the first article samples at 25%, 50%, and 75% points of completion of the production contract.

TABLE IV

NO.	CHARACTERISTIC	PER DWG OR SPEC	TEST PROCEDURE
501.	Humidity	3.3.2.5	4.6.2.5
502.	Fungus	3.3.2.6	4.6.2.6

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, Construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Interchangeability.- This test shall be performed utilizing an Electronic Enclosure Assembly, Dwg. F11737724 to meet the requirement of 3.3.1.

4.6.2 Environmental. - The assemblies shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the assemblies shall be subjected to the tests specified in 4.6.1 and 4.6.3.

4.6.2.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilites" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.3.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.6.2.2 Operational temperatures - The test shall be performed in accordance with Procedure I of MIL-F-13926 at the extreme temperatures specified in 3.3.2.2 utilizing the testing equipment indicated in 4.6.2.1. The assembly shall be exposed to each specified temperature for a period of two hours. At the end of each two hour period, and while still at the specified temperature, the assembly shall be subjected to the tests in 4.6.3. The assembly shall conform to the requirements of 3.3.2.2, 3.3.3.

4.6.2.3 Vibration.- This test shall be performed as per MIL-STD-810, Method 514.2, Procedure VIII, Curve V, Figure 514.2-6, resonance dwell not required. Subsequent to the vibration test the module shall meet the requirements of 3.3.3.

4.6.2.4 Shock. - This test shall be performed as per MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-1, Note 1. Subsequent to the shock test the module shall meet the requirements of 3.3.3.

4.6.2.5 Humidity. - This test shall be performed as per MIL-STD-810, Method 507.1, Procedure IV, Figure 507.1-3. Subsequent to the humidity test the module shall meet the requirements of 3.3.3.

4.6.2.6 Fungus. - This test shall be performed as per MIL-STD-810, Method 508.1, Procedure I. Subsequent to the fungus exposure the module shall meet the requirements of 3.3.3.

4.6.2.3 Performance - Operation.- Perform the tests 4.6.3.1 through 4.6.3.4 utilizing standard test equipment. When tested outside of the M55 LASER Trainer, the power supply shall always be mounted on a radiating type heat sink made of high thermal conductivity material of 1/4 inch thickness minimum, and whose minimum dimensions shall be five inches by two inches by one inch.

4.6.3.1 No load voltage. - Perform this test supplying the voltages as specified in 3.3.3.1 and measuring the output voltages using standard test equipment to conform to the requirements of 3.3.3.1.

4.6.3.2 Full load current.- Perform this test supplying the voltages specified in 3.3.3.2, and measuring the output current using standard test equipment to conform to requirements of 3.3.3.2.

4.6.3.3 Inhibit voltage test.- Perform this test supplying the voltages as specified in 3.3.3.3 and measuring the output voltages using standard test equipment to conform to the requirements of 3.3.3.3.

4.6.3.4 Flash mode test.- Perform this test supplying the voltages as specified in 3.3.3.4 and measuring the output voltages pulses, as shown in Figure I and Figure II using standard test equipment to conform to the requirements of 3.3.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11739861. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The high voltage power supply is a functional assembly of the LASER Trainer: M55. The assemblies major function is to provide regulated voltage.

6.2 Ordering data.- Procurement documents should specify the following :

(a) Title, number, and date of this specification.
(b) Applicable packaging data sheet number (see 5.1).
(c) Selection of applicable levels of preservation, packaging and packing.
(d) Applicable stock number.
(e) Requirement for First Article submission.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project Number 1240-A673

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-H-48408A(MU) High Voltage Power Supply: 11739861

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:25Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Phila., PA 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.